Section 4: Recommendations of Use: General Educational Purposes

After working with the hybrid model, users will discover that there are several ways to utilize the results to reach their objectives. The tool can accommodate instructors who need to tweak lesson plans based on performance points and administrators who want to determine reasonable pass/fail requirements (combined growth and proficiency) for employees. For them to rely on the tool, instructors and administrators must guarantee that the source of their data inputs is credible because low-quality inputs will result in low-quality outputs. Also, the performance criteria/rubrics must be consistent. Statistically speaking, there should be at least eight (8) credible data inputs accumulated over time to assess proficiency and growth at the beginning of any unit. For industry, this is equivalent to collecting work over a three-month span (E.G. probationary period).

A data input includes:

- Test scores from placement tests, diagnostic exams, midterm exams, final exams, take-home tests, and routine quizzes.
- The accumulation of comprehensive scores from homework, quizzes, worksheets, presentations, and in-class exercises across all skills for a given duration. The accumulation of scores can be successive or cumulative throughout the course offering period. The instructor is responsible for tracking accumulated scores from the beginning of the course to the end of the course.
- The accumulation of SMART metric scores, training scores (on-the-job/online/traditional), and internal performance scores is separated into theoretical components and practical components.

Normally, institutions and private/public instructors operate by one of the following schematics to obtain data inputs from a work period (WP) or an exam period (EP):

- WP for chapter 1 → EP 1 → WP for chapter 2 → EP 2 … → **Successive and Individual**
- WP for chapters 1-3 → EP 1 → WP for chapters 4-6 → EP 2 … → **Successive and Lumped**
- WP for chapters 1-3 → EP 1 → WP for chapters 1-6 → EP 2 … → **Cumulative**

Some instructors might also want a rule of thumb for creating tests and exercises for students with different interaction classes. Recall that:

→ Interaction Class 0 refers to the **failing** interaction class.
→ Interaction Class 1 refers to the **at-risk** interaction class.
→ Interaction Class 2 refers to the **acceptable** interaction class.
→ Interaction Class 3 refers to the **standard** interaction class.
→ Interaction Class 4 refers to the **advanced** interaction class.
→ Interaction Class 5 refers to the **exceptional** interaction class.

In general, the distribution of questions for students with different interaction classes should be:

- 95% acceptable level questions to 5% above level questions for Interaction Class 0.
- 90% acceptable level questions to 10% above level questions for Interaction Class 1.
- 65% acceptable level questions to 35% above level questions for Interaction Class 2.
- 60% acceptable level questions to 40% above level questions for Interaction Class 3.
- 35% acceptable level questions to 65% above level questions for Interaction Class 4.
- 30% acceptable level questions to 70% above level questions for Interaction Class 5.

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The distribution of questions reasonably challenges students at their respective levels regardless of whether the instructor chooses conventional testing (E.G. one test for all students) or disproportionate/personalized testing (E.G. separate tests for each student).

When making student growth projections, the instructor should use:

- The final three (3) performance points
- The first performance point, the median performance point, and the final performance point

When using human capital assessments for education, as a general rule, the user should score sorted receptive variables, sorted productive variables, and sorted acuity variables (if applicable) based on established benchmarks. Every time the totality of these variables is populated (daily, weekly, monthly, etc.), the totality should be collected in an assessment block.

Once the data has been fully obtained, the user can generate data analytics. Keep in mind that there are no data maturation periods, and outside work can be tracked using the negative PP IDs.

Handling Primary Learning & Focused Courses

Primary learning frameworks are designed to build foundational skills before a formal education can be delivered. Usually, these foundational skills include basic math, communication (speaking/writing), and reading; however, some students are exposed to vocational concepts and real-world phenomena at a rudimentary level. In order to comply with the robust Competency Accounting process of EvalzNow, the competencies linked to these foundational skills must be captured in different units as subjects for the same course.

As a simple example, suppose that a primary school teacher is teaching the subjects of reading, writing composition, and basic math to a group of K-3 students. The overall class, which is a unit, must be subdivided into smaller units to track the competencies of each subject. For the third sub-unit, basic math, it is easy to deduce that a T&P Unit Type is required, but what about reading and writing composition? What is the best way to handle reading and writing composition, especially for frameworks that force them to be assessed as separate items against well-established benchmarks?

The optimal way is to create two (2) sub-units, one for reading and one for writing composition, using the CI/CO Unit Type + weighted mathematical models and evaluating the benchmark criteria from a communicative perspective. The communicative perspective will be discussed further below.

SUB-UNIT #1: Reading → Unit Type – CI/CO + Receptive-Weighted Mathematical Model

The weight is on receptive skills because reading is a receptive skill.

The Unit Type, CI/CO, requires listening/reading scores as inputs and writing/speaking scores as outputs. By taking a communicative perspective, all inputs and all outputs can be accumulated, even though the course is targeting reading. Please see the points below.

- The student writes about a reading excerpt. The instructor accumulates a reading score and a writing score based on the process approach. The process approach ensures that the student's ideas address the reading and that the student's ideas are conveyed in a structured manner.
- The student speaks about a reading excerpt. The instructor accumulates a reading score and a speaking score based on the lesson plan. The lesson plan must ensure that the student's ideas address the reading and that the student's ideas articulate ideas in a structured manner.
- The student listens to various viewpoints on a reading excerpt before being quizzed on agreement and/or comprehension. The instructor accumulates reading scores and listening scores based on comprehension questions about and opinions on reading discussions.
- The student is challenged with read-aloud exercises and comprehensive questions about a reading excerpt. The instructor accumulates reading scores based on the quality of reading and direct comprehension questions.

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Once again, notice how all inputs and all outputs of the CI/CO can be accumulated using a communicative perspective, even though the course is targeting reading. Experienced instructors understand holistic learning well, so utilizing the communicative perspective for scoring should be straightforward because language skills must work together to achieve a given result. EvalzNow handles the rest with an emphasis on receptive skills.

SUB-UNIT #2: Writing Composition → Unit Type – CI/CO + Productive-Weighted Mathematical Model

The weight is on productive skills because writing is a productive skill.

The Unit Type, CI/CO, requires listening/reading scores as inputs and writing/speaking scores as outputs. By taking a communicative perspective, all inputs and all outputs can be accumulated even though the course is targeting writing – just like before. Please see the points below.

- The student writes about a reading excerpt. The instructor accumulates a writing score and a reading score based on the lesson plan.
- The student writes about a listening clip. The instructor accumulates a writing score and a listening score based on the lesson plan.
- The student speaks about the student's written works in speech. The instructor accumulates a writing score and a speaking score based on a presentation, a debate, a seminar, a conference, etc.
- The student is challenged with exercises targeting grammar, writing structure, and writing form. The instructor accumulates writing scores based on the quality of writing and direct syntax questions.

Once again, notice how all inputs and all outputs of the CI/CO can be accumulated using a communicative perspective, even though the course is targeting writing. EvalzNow handles the rest with an emphasis on productive skills.

Please note that the same perspective can be applied to public speaking courses.

Please note that we are providing the optimal approach assuming that an instructor is forced to assess reading and writing composition as separate items against well-established benchmarks. If this restriction is removed, then the instructor can assess reading and writing composition under the same CI/CO sub-unit due to the flexibility of EvalzNow. We do, however, expect the approach to subdividing units to be consistent from unit to unit, so the data sets can be comparable at the candidate level, the unit level, and the institutional level.

Handling Standardized Testing

The formulation of standardized tests, as well as their intended purpose altogether, has been intensely debated due to failing performance results, challenges in pervasive learning, poor teaching quality, time constraints, and questionable exam constructs. Proponents of standardized tests assert that these tests not only offer an objective measurement for gauging areas in need of improvement, but they also provide meaningful data to help marginalized groups excel in colleges or workplaces. Opponents, however, argue that standardized tests only determine which students are good at taking high-pressure tests (normally, with implications of test hacking rather than internalized learning) and that they do not address student progress, improved student performance, and future success – especially, from a diversity perspective. Some of these critics further argue that standardized tests are only useful for teacher evaluations.

Although the proponents and the opponents put forth sensible arguments, which might be valid or which might be invalid depending on a particular circumstance, toward standardized tests, both groups tend to skew away from the source of the problem – there is no hybrid, proficiency and growth, system outside of EvalzNow to rest upon in the present/absence of a standardized testing environment. We will further elaborate on how skewing away from the source of the problem causes issues for both sides and how EvalzNow fixes these issues.

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Coverage of Common Viewpoints on Standardized Testing

- Standardized tests used for grade failure/passage (i.e., district-wide, statewide, mid-terms, etc.) are proficiency tests because they pass the accountability to the student, and they are used as metrics to determine if a student can meet or can exceed a predefined benchmark or a baseline threshold, normally by demonstrating mastery of core concepts at a fixed point in time. Since these tests demonstrate mastery at fixed points in time, they can serve as objective measurements for gauging the understanding of a topic at a specific time; however, they cannot serve as objective measurements for gauging areas in need of improvement/growth because such measurements require mastery across a span of time and an engaging learning environment. During standardized testing, however, students cannot ask questions, cannot consult references/devices, and cannot interact with classmates.

EvalzNow addresses the difference between proficiency measurements and growth measurements. To circumvent time constraints and to prevent multiple benchmarks from being lumped in small question sets, we encourage educators to administer a proficiency test for each major benchmark and a comprehensive proficiency test for all benchmarks using the Cumulative schematic above. Once again, a proficiency test is a static measure that is only valid at a fixed point. It cannot be rolled over time-wise from performance point to performance point, and it cannot provide any feedback on how to perform better.

Additionally, EvalzNow holds students and teachers accountable during the learning process, and it can track tasks of both within the Competency Accounting process. To say that standardized tests are only useful for teacher evaluations is categorically false because teachers are deemed effective by student growth rates, and they are held accountable for improved student progress. A standardized test is associated with proficiency rather than growth, so the test should not be used for teacher evaluations.

- Standardized tests cannot provide meaningful data to help (non-)marginalized groups to excel in colleges or workplaces, unless they are provided as a supplement to the entire academic portfolio, which includes all proficiency metrics and all growth metrics at the very least. Put in another way, standardized tests cannot solely describe a student's academic/career potential.

EvalzNow provides an unlimited, academic portfolio comprised of proficiency metrics, hybrid metrics, growth metrics, utilization rates, attendance rates, neuroscience indicators, task logging, performance point mapping, and so much more that can be juxtaposed socioeconomically/demographically. This robust system should not be replaced by AP/IB/SAT/ACT/GMAT/GRE tests or similar, which are prone to test hacks, black-boxing answers, the fear of failure, and teaching-to-test approaches. Instead, these tests should be used as a supplement to the academic portfolio, which provides insight into a student's internalized knowledge capacity, including verbal reasoning and non-verbal reasoning, over time.

- Standardized tests determine not only which students are good at taking high-pressure tests but also which students are resourceful enough to reason out a solution to a problem when their knowledge capacity is limited. This problem-solving ability hinges on cognitive functions from neuroscience, and it can definitely be an indicator of future success depending on the construction of, the relevancy of, and the practicality behind the standardized tests; even though, the means to the end are unaccounted for.

EvalzNow was developed on the premise that students have different socioeconomics, diverse learning styles, and different approaches to challenges; therefore, the examiners must exercise due diligence to create culturally appropriate examinations/exercises of real-world value invariable of normed groups. Many examiners, for instance, create standardized tests that default to left-brained learners or uncommon, non-technical references with the assumption that present test-takers and future test-takers will be the same. Defaulting to left-brained learners stifles creativity and imagination, especially when removing collaborative projects/activities and utilizing uncommon, non-technical references alienates a large percentage of social groups whom are academically inclined despite being disadvantaged by unfamiliarity. Moreover, standardized tests are graded by machines that cannot process innovative solutions or creative responses to a problem. Not to mention, assuming that a normed group represents future test-takers, despite constantly evolving teaching/exam strategies, is unrealistic.

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To be clear, EvalzNow can either replace the need for standardized testing or it can absorb the results of the standardized test because EvalzNow tracks anything associated with knowledge capacity as a standalone system. This tracking includes learning experiences missed by standardized testing, such as leading group discussions, completing hands-on projects, and participating in innovative research.

Comments on Scoring and Credit

EvalzNow is a merit-based system, and it is an evidence-based system, so instructors should not input scores that are knocked down due to disciplinary reasons, such as behavioral penalties, perceived fairness, and attendance penalties. Such knockdowns will impugn upon the reliability of and the accuracy of the data submitted during analysis because they do not correlate hand-in-hand with the knowledge capacity. If an instructor, for example, knocks down a student's grade for multiple class disruptions after the student earned high marks on a group project, then the instructor might inadvertently misclassify the student into a lower interaction class (i.e., at-risk), leading to ill-conceived needs assessment actions and invalid knowledge capacity conclusions. We encourage users to link qualitative information, like disciplinary notes to performance points through observations or Candidate Marks, to avoid contaminating data pulls. At the end of a course, the instructor can analyze the qualitative data and the quantitative data to determine how to grade a student.

EvalzNow does not allow the attained score to exceed the maximum score for any assignment, so the user should rationally input scores for extra credit based on the following recommendations:

- For an extra credit assignment, which is given as additional practice or lesson reinforcement, the user can declare the delivery to be outside normal classroom instruction if the user does not want the assignment to impact the student's analytics profile. The performance point will have a negative ID.
- For an extra credit assignment, which is given as additional practice or lesson reinforcement, the user can declare the delivery to be a part of normal classroom instruction if the user wants the assignment to impact the student's analytics profile. The performance point will have a positive ID.
- Never add the extra credit score to the score of the original assignment because the score of the original assessment will be masked, and the sum of the scores might lead to a 100% exceedance error. Outside of risking an exceedance error, masked scores also make it difficult to hold instructors accountable during the learning process and to construct benchmark & PP ID mappings.
- Never replace the original assignment with an extra credit assignment, unless the extra credit assignment serves as an alternative to the "missed" original assignment or the student takes a zero-effort/no-effort mark on the original assignment with the expectation of replacement. This expectation can be based upon the instructor's permission, administrative policy, or a formal appeal. In either case, the student's performance should be logged into EvalzNow.

Educational Productivity

Another powerful function is the Butler Utility that evaluates the following:

1. The work effort accumulated from performance point to performance point
2. The effective work period in days vs. the actual tenure period in days
3. The productivity rates of a candidate

The work effort is only applicable to continuous performance point periods. The effort calculation is a function of the total number of performance points, productivity rates, effective work periods, tenured time, and the amount of work captured in a given performance point.

The effective work period scales the actual tenure period by a magnification factor based on the anticipated work completion times set by a Scope of Work (SoW), assigned tasks, a schedule, or an official agreement, and the temporal opportunity to complete work.

For educational institutions, the productivity rates of students should be between 85% and 95% at least. We recommend using the Unit Start Date as a reference for the Butler Utility.

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We understand that educators require seamless, user-friendly interfaces with small learning curves, so the EvalzNow application has been oversimplified with guidance text and visual icons. There are no hidden menus or special right-click features. Moreover, the user only has to follow these simple steps:

1. Use the +Staff Control menu to add staff data to the database.
2. Use the +Candidate Control menu to add candidate* data to the database.
3. Use the +Unit Control menu to open courses based on grading scale definitions, unit benchmarks, a unit type, and a signature profile.
4. Use the +Unit Control menu to assign staff and candidates to the newly created unit.
5. Use the +Unit Control menu to open assessment blocks for candidates.
6. Use the Insert Scores module to add scores to the assessment blocks.
7. Use the Initiate Analysis module to generate data analytics and tabulations at the candidate/unit level.
8. Add qualitative details based on interpretations of results at the candidate/unit level.
9. Extract raw data at the candidate/unit level.
10. Generate reports at the candidate/unit level.

A candidate in EvalzNow refers to any person under assessment. This person can be a student, an administrator, an instructor, an educational consultant, and so forth.

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